

No. of Functional Features Included in the Solution

Date	01 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Collects soil and environmental parameters from users	Frontend UI	Done	High
2	Data Preprocessing	Cleans and prepares input data for prediction	Data Processing	Done	High
3	Crop Recommendation Model	Predicts the most suitable crop using ML algorithms	Machine Learning	Done	High
4	Model Evaluation	Compares model performance and selects the best model	Machine Learning	Done	Medium
5	Flask Backend Integration	Connects frontend with trained ML model	Backend	Done	High
6	Prediction Result Display	Displays recommended crop to the user	Frontend UI	Done	Medium
7	Input Validation	Validates user-entered values before prediction	Backend	Done	Medium

8	Model Persistence (.pkl)	Saves and loads trained model for predictions	Backend / ML	Done	Medium
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Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Result Display
2	Backend / Logic	3	Flask Integration, Input Validation, Prediction Processing
3	Database / Storage	1	Model Persistence using Pickle (.pkl)
4	API / Integration	0	Not Applicable
5	Security / Authentication	0	Not Implemented